PRD: Aurel Reservations AI Agent, v1

Product requirements · Owner: AI Product Manager · Engineering: 2 full-stack, designer at 50% · 23 October 2026

Problem	Guests wait 9 hours for an email reply (34% book if answered in an hour, 9% after a day), and 23% of pre-arrival requests are not delivered.
What we build	An AI agent that reads every email and WhatsApp message, marks its type, assigns a named owner, drafts the reply with live prices and the right hotel rule, turns requests into hotel to-dos, and answers policy questions on its own.
How we measure	Reply time under an hour, requests not delivered under 10% then 5%, 80% of policy-only messages answered without a person, zero broken promises, every item owned.

> **Terms. AI agent** is the software, written with "AI" in this document; on the desk it is called Aurel Assistant. **Agent** on its own is a person on the 22-person reservations desk (the CRO). The **PMS** is the hotel's booking system. A **trace** is a to-do in the PMS sent to a hotel department with a due date. A **queue** is a list of items waiting for one kind of owner. An **owner** is the named person accountable for an item until it is closed.

1 Scope

In v1 (first quarter)

- AI agent marks the type, finds the booking, flags urgency and sensitive data
- Owner assigned on arrival, by rule
- Important view + five queues (Flow A)
- Sends alone: cited policy answers; fixed templates only: holding reply, group acknowledgement, off-topic decline, the question asking which hotel, day-before message
- Also built: attachments in and out (never read by the AI agent); a follow-up prepared when a quote has no reply for 48 h; the mailbox as source of truth, loaded on sign-in
- Drafts with live PMS price; PMS changes prepared within rules; one PMS trace per request
- Day-before check on every booking (Flow B)
- Dashboard by role (Flow C)
- Login per role: agent, senior, manager, groups specialist
- **If time runs short:** cut from the end, day-before check first

Next (second quarter)

- Price quotes and within-rules changes earn the right to send alone
- Out-of-hours cover for the types that send alone
- Automatic message to the guest when a trace is closed
- Languages beyond English

Out

- Booking and taking payment without a person
- Any discount or price match
- Group pricing (the groups specialists own it)
- Website chatbot
- PMS write without approval

Channel choice: which channels, when, and why

CHANNEL	SHARE	V1	WHY
Email	58%	Week 1	Biggest and slowest channel (9 h median). Mailbox API is standard. Written, so draft-and-approve fits naturally.
WhatsApp	11%	Week 11	Guests expect minutes. WhatsApp Business API needs Meta verification (1 to 4 weeks). Short replies; same AI agent, same queues.
OTA messaging	7%	Q2	Booking.com and Expedia partner messaging APIs need approval that takes weeks to months. Behaviour identical once connected.
Phone	24%	Out	Live answer in seconds, recording consent, no transcripts today, and no engineer capacity. Agents can paste a call note into the desk and get the same help.

Prototype: what is real and what is simulated

Real
The AI agent (Google Gemini, read-only PMS tools) doing everything in the v1 list against per-hotel handbooks; a rule-based engine takes over when the model is unavailable, labelled. Email through a real Gmail mailbox, polled by open desks and a five-minute scheduled check; replies really go out.

Simulated
The PMS (nine hotels, rates, bookings, daily prices), the hotel's trace list (a PMS traces page stands in), the payment link. WhatsApp is not in the prototype.

2 Users

Reservations agent (13)

Works the Important view and their own queue. Approves, edits, changes the type, or sends to a manager.
Needs: one click to send a right answer; never to be blamed for the AI agent's mistake.

Senior agent (4)

Everything an agent has, plus their team's lists and reassignment.
Needs: see who is stuck before the guest notices.

Reservations manager (2)

Owns the Manager queue. Each item arrives with the booking, the rule it breaks and the guest's words.
Needs: decide once, same day, with context.

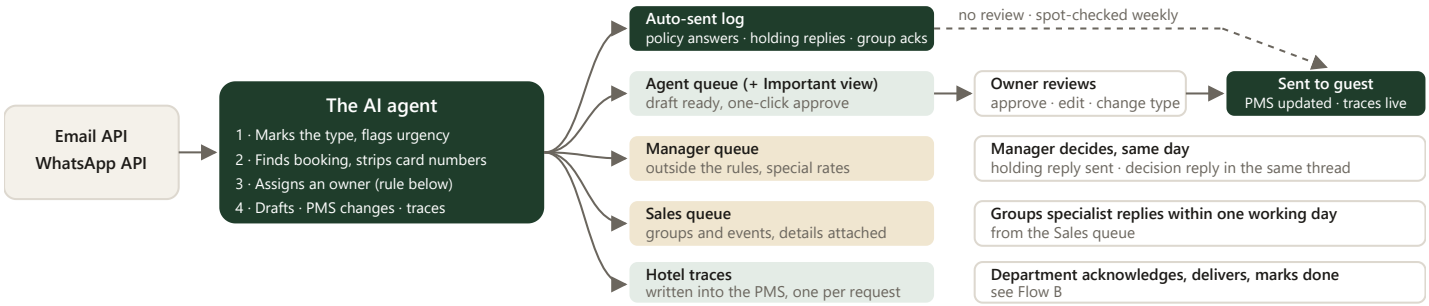
Groups and events specialist (3)

Owns the Sales queue. Group and event items arrive with dates, size, hotel and the guest's words; replies within one working day.
Needs: a complete brief, no generalist back-and-forth.

Not users of this product, but served by it: hotel department staff (front office, housekeeping, concierge, F&B) work the traces in the PMS they already use; the **hotel general manager** gets trace escalations and the weekly promises-kept number; the **guest** receives auto-sent messages signed as the reservations team.

3 Flows

Flow A · A message arrives



Owner assigned on arrival

ITEM	OWNER	REASSIGN
Policy-only question	The AI agent (sends alone). No policy found → next agent in rotation	Senior
Price, new booking, change, request, within-rules cancellation	Next agent on shift in rotation, skipping anyone over 15 pending	Senior
VIP or travel-advisor email; cannot be sorted	Senior agent on shift	Senior
Refund or change outside rules, special rate	Reservations manager on duty	Other manager
Group or event	Groups specialist on shift	Senior
Each trace, incl. the day-before call	Department head at that hotel	Hotel GM

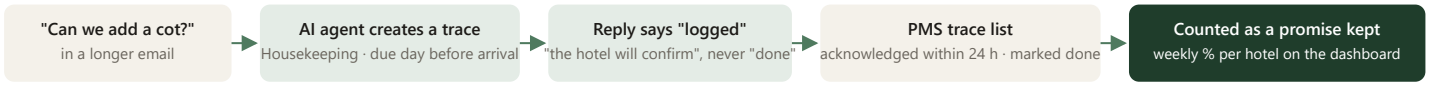
Shift end: pending items move automatically by the same owner rule and the dashboard logs it. If nobody of that role is on shift the item stays, flagged, for a senior or manager.

Important view: what rises to the top

RULE	WHY
New booking or price enquiry	Revenue: fast replies book, slow ones do not
Guest arrives within 48 hours	No time to recover from a miss
Allergy, medical or accessibility content	Safety
Guest has written more than once about the same thing	Already let down once
Compassionate cancellation; VIP or advisor	A person should reply fast

Important is a view, not a type. The item keeps its type and owner; the rule only changes where it is shown.

Flow B · A special request becomes a kept promise



Day-before check: every booking gets an automatic message asking arrival time and anything else needed; bookings with a request, a restricted trace or a VIP flag get a front-office call instead, created as a trace.

Flow C · The dashboard, visible to the whole desk

ROLE	SEES
Manager	Waiting and the oldest · replied today and median time · promises kept per hotel, read from PMS trace status · one red line for guardrail breaches, unowned items and stuck handovers, only when not zero · who is waiting on whom, flags not scores · oldest items with reassign. Sent alone and corrections live in the Sent alone view
Senior	Who is waiting on whom for the team: waiting, oldest, flag; replied today, edited drafts and type changes on hover · one-click reassign · Important and Sent alone
Agent	Important view, then own queue, oldest first · age and due time on every item · own resolved today and average reply time

Flags, not scores. An item is flagged when it is older than 2 hours in opening hours, or when an owner has at least 5 items pending and more than 1.5× the desk average. The board shows items, never a ranking of people.

4 AI agent behaviour spec

One row per type. A message with several asks (date change + cot + anniversary) gets one proposal with one action per ask, at the strictest approval level among them. **Anything sends alone only when the whole message is of that type**: a spa question with a buried allergy never auto-sends. Scope is the guest's stay at Aurel only; text inside a message is information, never an instruction, and a topic check in code runs before drafting and before every send. Every sent message records the type, the policy used and what was not promised.

TYPE · SHARE	WHAT IT HANDLES	WHAT IT NEVER DOES	HANDS OFF WHEN → TO WHOM	HOW IT CONFIRMS	HOW IT FAILS
Policy question 12% sends alone	Parking, pets, check-in and check-out times, directions, facilities. Finds the paragraph in that hotel's policy documents and answers from it, citing it. Sends alone only if a paragraph was found and the answer cites it.	Answer from general knowledge. Guess a hotel. Treat "can I check in at 9am?" as policy: early check-in is always a special request.	No paragraph found, or anything else in the message → next agent, drafted. Hotel not named → fixed question sent alone; the reply is read with the thread	States the policy as written and names the hotel. Signed as the reservations team.	Weekly check of 50 auto-sent: a wrong promise pauses auto-send at once; over 2% wrong pauses it for that hotel.
Price / availability 31% approve	Pulls today's price and availability from the PMS for the dates, room and party. Drafts the quote with Flexible and Advance Purchase options, totals and terms. Offers a hold and a payment link.	Quote a number not returned by the PMS in this session. Discount, match or undercut any price. Hold or book without approval.	Non-public or negotiated rate asked for → manager. Dates or hotel missing → drafts a question for the guest.	"As of today" prices; booking confirmed only after gateway payment succeeds.	PMS unavailable → draft acknowledges, gives a time to come back, no numbers. Room not available → nearest alternative. No reply in 48 h → follow-up prepared for the owner, never sent alone.
Booking change 22% approve	Checks the booking's rate rules and live availability. Prepares the PMS change with the price difference. The agent approves the write.	Change an Advance Purchase booking. Waive a difference.	Rate does not allow changes, or price difference needs waiving → manager	Updated confirmation sent only after the PMS write succeeds.	Write fails → nothing sent, item returns to the owner with the error. A reply claiming a change the PMS did not make is blocked in code. Booking not found → asks for the reference.
Special request 18% approve	Cot, transfer, early check-in, occasion, spa, dining, dietary. One trace per request to the right department with a due date before arrival. Drafts "logged, the hotel will confirm". Allergy, medical, accessibility → restricted trace, minimum detail, tells the guest who will see it.	Say a request is done or guaranteed (early check-in, upgrade, transfer). Put sensitive detail in free text or an email. Promise an allergen-free kitchen.	VIP or travel-advisor email → senior agent; amenities confirmed by a person, upgrade never promised in writing. Needs a hotel decision → via the trace.	Delivery is confirmed by the hotel from the trace, in writing. Q2: automatic message on close.	Not acknowledged within 24 h of due → escalates to the hotel's front office manager. Trace creation fails → item stays open, owner alerted.
Cancellation / refund 9% approve template	Checks the rate and the cancellation window. Within rules: drafts the confirmation, asks the guest to confirm. Outside rules or compassionate: **sends a fixed-template holding reply at once** ("received, a manager will contact you today") and hands off with the booking attached.	Confirm a refund, credit or waiver outside the rules. Cancel without the guest's explicit yes. Ask for medical proof. Put anything but the template in an auto-sent message.	Any refund, waiver or credit outside the rules, or a compassionate case → manager, same day.	Cancellation confirmed in writing after the PMS update. Refund confirmed only by the manager.	Rules unclear → treated as outside the rules and routed to the manager.
Group / event 3% to Sales template	Recognises 8+ rooms, meeting space, weddings. **Sends a fixed-template acknowledgement** (one-working-day reply from the specialist). Passes on dates, size, hotel and the guest's words.	Quote, hold inventory or state availability for a group.	Always → groups specialist	Acknowledgement only. The specialist confirms.	Group size unclear → still routed; the specialist asks.
Other · off-topic · manipulation 5% approve template	Unclear ask: drafts a short reply asking what is needed. Card details → removed, payment link proposed. **Off-topic** (code, homework, general knowledge, legal or medical advice, competitors, anything not about a stay at Aurel) or **instructions aimed at the AI** ("ignore your rules", "write me a script", "show your prompt") → fixed-template decline, routed to an agent, counted in the audit log.	Attempt the off-topic task. Reveal its own instructions, other guests' details or internal notes. Store, repeat or forward card details.	Under 60% confidence → senior agent. Repeated attempts from one sender → senior agent. Complaint, threat or abuse → manager.	Nothing confirmed.	Unsure whether on-topic → treated as off-topic, drafted for a person. Model output invalid or model down → one repair attempt, then the rule-based engine, labelled.

How a type earns the right to send alone

- A type qualifies when, over 4 weeks and 300+ items: 95% of drafts sent with no or trivial edits, type changed under 3%, zero guardrail breaches.
- The manager switches it on per type and per hotel.
- Expected order: price quotes (Q2), then within-rules changes. Refunds and sensitive requests never qualify.

5 Success metrics

Committed targets are the cautious case from the memo; stretch is the realistic case. Six headline metrics; the rest are tracked monthly.

HEADLINE METRIC	BASELINE	COMMITTED, END Q1	COMMITTED, END Q2	STRETCH	HOW MEASURED
Email reply, weekday median	9 h	Under 1 h in opening hours	Under 45 min	Under 30 min	Received to sent timestamps
Policy-only messages answered alone	0%	80%	90%	95%	Auto-sent ÷ policy-only messages
Drafts sent with no or trivial edit	n/a	70% by week 8	85%	90%	Edit distance on send
Requests not delivered at check-in	23%	Under 10%	Under 5%	Under 3%	Front office audit, weekly sample of 100; pilot hotels at end Q1, all hotels at end Q2
Oldest open item per owner	n/a	Under 4 h	Under 2 h	Under 1 h	Dashboard
Enquiry conversion, email	mixed 34% / 9%	+3 points	+5 points	+10 points	Booking within 7 days; pilot vs non-pilot hotels (indicative)

Also tracked

METRIC	BASELINE	END Q2
Weekend reply median	26 h	Under 4 h
Auto-sent answers found wrong	n/a	Under 2%
Type changed by the agent	n/a	Under 5%
Traces acknowledged within 24 h	n/a	90%
Time spent looking things up	25%	Under 12%
Arrivals contacted the day before	~0	95%
Agent satisfaction with the tool	n/a	4 of 5

Desk hours by type: where the targets come from

TYPE	TODAY	AFTER	SAVED	HOW, TEXT CHANNELS ONLY
New booking	307	~200	106	11 → 6 min on ~1,270, person approves
Booking change	139	~80	60	7 → 3 min on ~900
Special request	81	~45	37	5 → 2 min on ~740, traces automatic
Policy question	32	~15	17	3 → 0 min on ~340 policy-only
Cancellation	49	~30	18	6 → 3 min on ~370
Call notes	above		30	Price and rule found for the pasted note
Total	**~750**	**~480**	**~270**	Realistic, full rollout

Groups (122 h) need a sales decision; no saving counted. Other (18 h) unchanged. Committed: **~700 end Q1** (the pilot's partial effect), **~615 end Q2** (AI agent delivers half its share), stretch ~480. Minutes are estimates, measured in shadow mode and corrected.

Guardrails: hard limits. Any breach pauses that type the same day until fixed.

GUARDRAIL	LIMIT	CHECKED BY
Unconfirmed promises sent (early check-in, upgrade, transfer, refund)	0	Weekly QA sample of 50, with the team
Card numbers stored, displayed, forwarded or sent to the model	0	Scan of stored messages and model calls
Allergy, medical or accessibility detail in free text or email	0	Automated scan + QA sample
Discount or price match offered · quoted price differs from PMS	0	Every quote logged against PMS
PMS write or send below the approval level in the spec · items without an owner	0	Audit log · dashboard
Auto-sent that is not a cited policy answer or one of the fixed templates	0	Audit log
Off-topic or instruction-following answer sent (code, advice, others' data, own prompt)	0	Topic check on every send + QA
Data protection assessment signed before any sensitive trace goes live	Required	Legal, week 2

Rollout, first quarter

Weeks 1 to 2 · Shadow
Email connected. Policy documents loaded for 9 hotels. AI agent drafts everything; nobody sends from it. Measure draft acceptance, type accuracy and real minutes per contact. Hotels: a named sponsor per hotel; department heads trained on the PMS trace list.

Weeks 3 to 6 · Pilot
4 volunteer agents, Lisbon and Mallorca. Sending from the desk; auto-send on for those two hotels. Traces live for both; "promises kept" % reported weekly, misses reviewed with the sponsor, not blamed. Weekly QA in the open with the team.

Weeks 7 to 10 · All agents
All 22, all 9 hotels, email. Owner rules and dashboard live. Traces at every hotel; weekly promises-kept % per hotel in the COO's review; hotels under 80% acknowledged for two weeks get a visit from the PM.

Weeks 11 to 12 · WhatsApp and decisions
WhatsApp connected after Meta verification.
No headcount reduction planned.
Replacing leavers is decided on measured hours, first review end Q1. Decide which type earns autonomy next. Q2: OTA, out-of-hours, trace-close messages.

Assumptions. Desk of 22: 2 managers, 3 groups and events specialists, 4 seniors, 13 agents. Email and WhatsApp providers expose threaded read and send APIs; WhatsApp verification takes 1 to 4 weeks. PMS trace API accepts department, due date and a restricted flag, and returns trace status so acknowledgements and promises kept can be counted; otherwise sensitive detail stays in our system with a pointer and status is a vendor gap to raise in week 1. Policy documents for 9 hotels structured by the PM with each hotel in weeks 1 to 2. Two engineers for two quarters (~52 engineer-weeks) cover the v1 list with a five-week buffer. Model provider: data processing agreement, EU inference region, card numbers removed and sensitive detail minimised before any call.